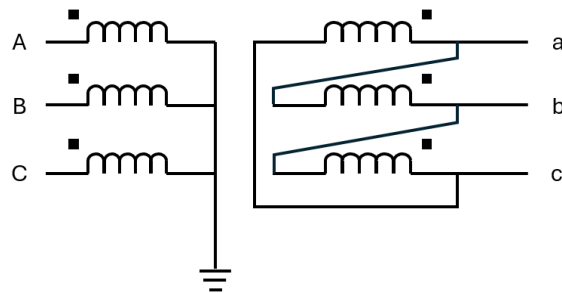


ELEC4142 Power System Protection and Switchgear

Transformer Protection – Example

1. a) [Rotation Matrix]

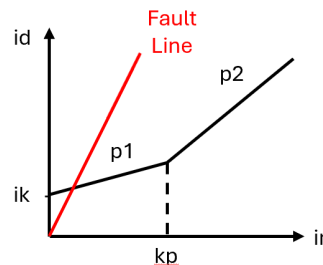
Determine the rotation matrix for the following ICT connection.



- b) [Current Distribution in BD/REF Winding under Fault]

A 80MVA 132/33kV Ynd11 bulk supply transformer is earthed solidly on the star. Biased differential scheme is employed to provide protection for the transformer.

- Illustrated, with a three-phase wiring diagram, how biased differential and restricted earth fault protection can be applied for the protection of the power transformer.
 - Assume 1A secondary CT rating, design suitable current transformer ratio.
 - Determine the fault current distribution in the power transformer, CTs, ICTs and relays for a phase-to-phase external through fault current of 6270A on the 33kV.
- c) [Bias Differential Setting and Injection Test]
- Given a 50MVA 132/11kV Yyn0 transformer with impedance 27% and tap range from +10.5% to -21%. The HV line CT is in 1/400T, Class X; and the LV line CT is in 1/1600T, Class X.
- Determine the HV and LV ICT ratio to achieve unit load at relay side.
 - Determine the largest differential current at any taps with maximum CT error of $\pm 5\%$.
 - Determine the biased differential setting for the transformer.



- Determine the single end secondary injection with phase-to-phase and phase-to-earth configuration to activate biased differential element.
- d) [Conceptual Questions]
- Discuss the differences between classical E&M relay and digital relays on how to maintain stability with inrush and provide sufficient sensitivity for earth fault.
 - Discuss the working principles of directional element and the necessity of directional element in protection scheme for parallel transformers.
 - Discuss the working principles for Buchholz relay and suggest a fast-operating circuit design for Buchholz surge trip (BST) with adequate trip contacts.
 - Discuss the required pre-commissioning and commissioning tests for newly installed RMU-fed transformers.
 - Suggest one inrush detection algorithm and one external fault detector (EFD) to maintain stability against inrush.